



New data on the evolution of the Cretan spiny mouse, *Acomys minous* (Rodentia: Murinae), shed light on the phylogenetic relationships in the *cahirinus* group

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The karyotype of the Cretan spiny mouse *Acomys minous* was examined with chromosome banding techniques in 53 individuals from 12 localities of Crete, aiming to gain a more detailed knowledge on the chromosomal constitution and variability of its natural populations. We found that it consists of three Robertsonian (Rb) populations with $2n = 38$, $2n = 40$ and $2n = 42$, respectively, the last one being reported for the first time, and with stable fundamental number (FNa = 66, FN = 68). The G-banding pattern proves that the Rb populations are closely linked phylogenetically by the many common Rb fusions and the lack of monobrachial homologies. In addition, they appear to freely mate at their contact areas, producing viable and fertile hybrids. No other type of chromosomal rearrangement appears to have played part in the chromosomal evolution of this species, at least in the recent past, as indicated also by the study of the telomeric sequences. Heterochromatin appears to be restricted to the pericentromeric position of all acrocentric and most biarmed autosomes, as well as of the X chromosome, whereas the Y chromosome is uniformly, yet faintly heterochromatic. Chromosome banding comparison of the karyotypes in *A. minous* with those of the other species in the *cahirinus* group (i.e. *Acomys cahirinus*, *Acomys cilicicus*, and *Acomys nesiotus*) proves their very close phylogenetic relationship, further reinforced by the study of the cytochrome *b* sequences, and that *A. minous* possesses the ancestral karyotype of the group. It is suggested that at least two of the karyotypes that characterize *A. minous* today, pre-existed in North Africa before it colonized Crete and that the specific status of the four members in the *cahirinus* group may need to be revisited. © 2011 The Linnean Society of London, *Biological Journal of the Linnean Society*, 2011, 102, 498–509.

ADDITIONAL KEYWORDS: colonization – cytochrome *b* – geographic isolation – heterochromatin – karyotype – Robertsonian fusion – telomeric sequences.

INTRODUCTION

The spiny mice of the genus *Acomys* have a widespread distribution over Africa, the Arabian

Peninsula, Anatolia (Turkey), as well as the islands of Crete and Cyprus in the Mediterranean region. Because of the absence of undisputable diagnostic markers, the taxonomy of this genus, still under clarification, has changed several times and a varying number of taxa has occasionally been suggested as valid (Bates, 1994; Musser & Carleton, 2005). The

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existing karyological studies on the genus *Acomys* have revealed a significant variation in the diploid chromosome number, ranging from $2n = 36$ to $2n = 68$ and a relatively lower variation in the fundamental number (FN = 66–74), thus indicating a predominance of Robertsonian (Rb) fusions in the karyotypic evolution of this genus (karyotypic orthoselection) (Matthey, 1963; Wahrman & Goitein, 1972; Volobouev, Tranier & Dutrillaux, 1991; Sokolov *et al.*, 1993; Denys *et al.*, 1994; Volobouev *et al.*, 1996a, 2002; Kunze *et al.*, 1999a; Nicolas *et al.*, 2009).

The Cretan endemic, *Acomys minous*, belongs to a group of closely related taxa (i.e. the *cahirinus* group), along with *Acomys nesiotis*, distributed in Cyprus, *Acomys cilicicus* from Anatolia (known only from around its type locality), and *Acomys cahirinus*, with a relatively wider African distribution (Musser & Carleton, 2005). *Acomys minous* was initially considered to be represented by two chromosomal populations (i.e. Race I with $2n = 38$, FN = 66 and Race II with $2n = 40$, FN = 68; *sensu* Matthey, 1963). The existence of a population with $2n = 38$ was also later verified by Kunze *et al.* (1999a), albeit with FN = 68. The remaining three species of the group appear to present quite similar karyotypes to *A. minous*, showing complete homology for up to fifteen biarmed and one acrocentric pair of autosomes, thus indicating a strong phylogenetic relationship among them. Specifically, *A. nesiotis*, is also characterized by $2n = 38$, FN = 68, whereas *A. cahirinus* and *A. cilicicus* are both characterized by $2n = 36$, FN = 68 (El Ashmawy, 1990; Volobouev *et al.*, 1991; Macholán *et al.*, 1995; Volobouev *et al.*, 1996a; Volobouev, Gautun & Tranier, 1996b; Zima *et al.*, 1999; Kunze *et al.*, 1999a). It is expected that additional karyological investigation, both at intra- and interspecific levels, will help clarify further the phylogenetic evolution of the *cahirinus* group, especially because the results from molecular [cytochrome *b* (*cytb*)] studies are not considered sufficient on their own for drawing definite conclusions (Barome, Monnerot & Gautun, 1998, 2000; Barome *et al.*, 2001a). However, with regard to *A. minous*, it has been possible to detect the existence of two distinct maternal mitochondrial (mt)DNA lineages (Group 'A' and Group 'B'), even though no attempt has been made to correlate the mtDNA variation with the chromosomal variation in this species. Group 'A' of *A. minous* appears to cluster with *A. nesiotis* and *A. cahirinus*, whereas Group 'B' of *A. minous* with *A. cilicicus* (Barome *et al.*, 2001a).

Other species with low diploid chromosome numbers in the genus *Acomys* have been placed in the *cahirinus-dimidiatus* complex, which is characterized by a low interspecific morphological differentiation (Petter, 1983; Volobouev *et al.*, 1991; but see also Denys *et al.*, 1994): *Acomys dimidiatus* with $2n = 38$,

FN = 70 and *A. airensis*, recently proposed as a junior synonym of *Acomys chudeaui* by Nicolas *et al.* (2009), with $2n = 40$ –46, FN = 68. As a result of the study of chromosome banding patterns, it has now been established that the karyotypes of *A. cahirinus*, *A. dimidiatus*, and *A. chudeaui*, although apparently derived from a common ancestral form with $2n = \text{FN} = 68$ (*Acomys* sp.), are clearly distinguished because of the independent accumulation of successive Robertsonian fusions in allopatry, resulting in extensive monobrachial homologies among them (Volobouev *et al.*, 1991, 1996a, b, 2007; Kunze *et al.*, 1999a). This discrimination is also corroborated by molecular (Barome *et al.*, 2001a; Nicolas *et al.*, 2009) and morphological studies (Denys *et al.*, 1994).

On the basis of all the above, it appears that, within the *cahirinus-dimidiatus* complex, the most problematic issue appears to be the resolution of the phylogenetic relationships among the taxa of the *cahirinus* group. Therefore, in the context of the present study, it was deemed necessary to further investigate the degree of chromosomal variation in populations of the chromosomally diverse *A. minous*, with the use of different karyological approaches (study of G- and C-banding patterns, meiosis, and telomeric repeats), aiming, with the help of *cytb* studies, to examine the possibility of correlation between mtDNA and chromosomal variation in this species, to better describe its phylogenetic position within the *cahirinus* group and to suggest potential colonization routes to Crete for this taxon.

MATERIAL AND METHODS

A total of 53 individuals (34 males and 19 females) from natural populations of *Acomys minous* were captured from 12 different localities of Crete (Fig. 1, Table 1). The individuals were brought to the laboratory, where they were subjected to karyological and molecular studies.

KARYOLOGICAL STUDY

Mitotic chromosome preparations were obtained directly from bone marrow (Hsu & Patton, 1969), after subcutaneous injection of yeast stimulation solution 72 and 24 h and intraperitoneal injection of 0.025% colchicine solution 45 min before sacrifice (Lee & Elder, 1980). Meiotic slide preparations were obtained from testicular material of some males, *sensu* Evans, Breckon & Ford (1964), and stained with Giemsa. Banding pattern analyses were accomplished, using the G- and C-banding staining techniques, *sensu* Seabright (1971) and Sumner (1972), respectively, after some modifications.

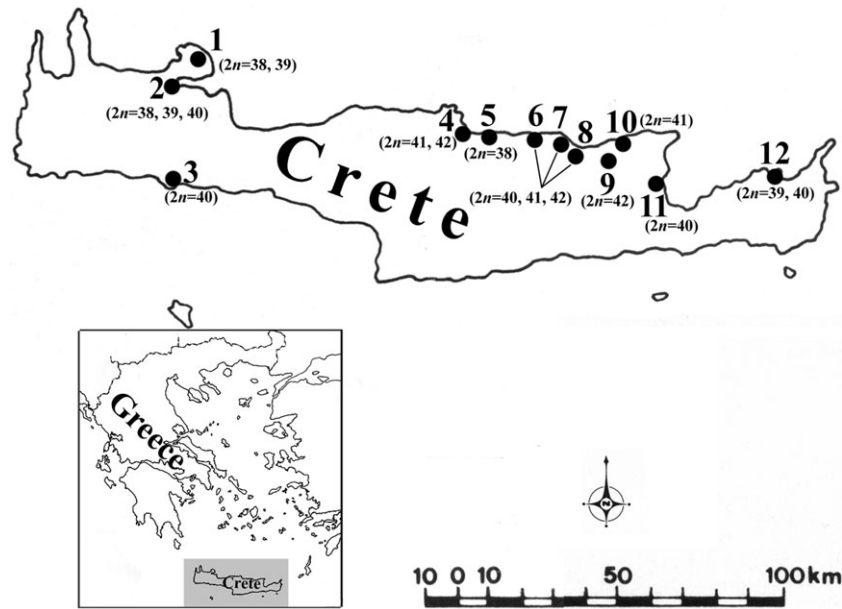


Figure 1. Map of Crete, showing the collection localities and the diploid chromosome numbers of the *Acomys minous* sample used in this study.

Table 1. Sampling localities of *Acomys minous* from Crete, number of studied animals, and their karyological characteristics

		Number of individuals													
		$2n = 42$		$2n = 41$ Rb(17.19)*		$2n = 41$ Rb(15.16)*		$2n = 40$ Rb(17.19)		$2n = 39$ Rb(17.19) Rb(15.16)*		$2n = 38$ Rb(17.19) Rb(15.16)			
(Number)	locality	♂	♀	♂	♀	♂	♀	♂	♀	♂	♀	♂	♀	Total	
Crete															
Prefecture of Chania															
(1)	Akrotiri	—	—	—	—	—	—	—	—	2	1	2	—	5	
(2)	Souda	—	—	—	—	—	—	1	—	1	1	—	1	4	
(3)	Anopoli	—	—	—	—	—	—	4	—	—	—	—	—	4	
Prefecture of Iraklio															
(4)	Linoperamata*	—	2	—	1	1	1	—	—	—	—	—	—	5	
(5)	Iraklio vicinity	—	—	—	—	—	—	—	—	—	—	1	—	1	
(6)	Chani Kokkini	—	1	2	2	—	—	2	—	—	—	—	—	7	
(7)	Piskopiano	1	—	2	—	—	—	2	2	—	—	—	—	7	
(8)	Stalida-Mochos	1	1	1	—	—	—	2	—	—	—	—	—	5	
Prefecture of Lasithi															
(9)	Vrahasi	2	—	—	—	—	—	—	—	—	—	—	—	2	
(10)	Tsampi	—	—	—	1	—	—	—	—	—	—	—	—	1	
(11)	Ag. Nikolaos	—	—	—	—	—	—	—	2	—	—	—	—	2	
(12)	Sitia	—	—	—	—	—	—	4	2	3	1	—	—	10	
		8		9		2		21		9		4		53	

Locality numbers correspond to Fig. 1. *Rb chromosomes in heterozygosity. The use of bold is intended to clarify total and grand total numbers.

To determine the location of telomeric repeats, the telomeric sequence probe was prepared and labelled in a single polymerase chain reaction (PCR) without template DNA, by using two pairs of primers, (TTAGGG)₅ and (CCCTAAA)₅, in the presence of dUTP-biotin (Roche). The relevant protocol has been described previously (Ijdo *et al.*, 1991). Fluorescent *in situ* hybridization was performed, as outlined elsewhere (Fernández *et al.*, 2001), with slight modifications. Slides were counterstained with 4'-6-diamidino-2-phenylindole in Vectashield (Vector Laboratories) and several metaphases were studied.

DNA SEQUENCING

Total DNA extraction from ethanol preserved liver tissue was carried out in 20 representative specimens, using the Macherey-Nagel Tissue kit, in accordance with the manufacturer's protocol and examined through agarose gel electrophoresis. The *cytb* gene was amplified by PCR, using an Eppendorf Mastercycler. PCR reactions were carried out in 50 µL volumes (5 units HyTest Ltd Taq polymerase, 5 µL 10 × HyTest Ltd. PCR buffer, 0.2 mM dNTPs, 2 mM MgCl₂, approximately 100–200 ng template DNA, and 10 pmol of each primer, filled to 50 µL with water). Amplification of the entire *cytb* gene was performed by using one primer set (ND6-2 and thr3), as previously described by Barome *et al.* (1998). The implemented thermocycler program was: one preliminary denaturation step at 93 °C for 3 min followed by 35 PCR cycles, each consisting of strand denaturation at 93 °C (1 min), annealing at 55 °C (1 min), and primer extension at 72 °C (1.5 min). A final extension step was performed at 72 °C for 5 min. The PCR products were purified, using the Nucleospin extract II kit (Macherey-Nagel), in accordance with the manufacturer's protocol. Individual sequences were determined via automated sequencing of both strands of *cytb* gene, provided by the VBC and MacroGen Company. The primers in the sequencing reactions were the same as those employed in the amplification procedure. The obtained sequences have been deposited to GenBank under accession numbers GU046546–GU046553 and HM63803–HM63814. Previously published sequences from *A. minous*, *A. nesiotes*, *A. cilicicus*, *A. cahirinus*, *A. chudeaui* and *A. dimidiatus* were also used for the phylogenetic study, whereas *A. ignitus* was used as outgroup (Barome *et al.*, 1998, 2001a, b; Nicolas *et al.*, 2009). Amino acid sequences were translated from nucleotide sequences, applying the vertebrate mitochondrial genetic code.

DATA AND PHYLOGENETIC ANALYSIS

Multiple-sequence alignments were done with CLUSTALW2 (Thompson, Higgins & Gibson, 1994) of

the ClustalW2 Service at the European Bioinformatics Institute (<http://www.ebi.ac.uk/Tools/msa/clustalw2/>), using the default parameters. The computer-generated alignment was further adjusted manually. Additionally, the alignment of the *cytb* data set was verified against *cytb* sequences of other *Acomys* specimens available in GenBank. The sequences produced were unambiguously aligned to the retrieved *cytb* sequences, whereas no gaps and/or stop codons were present. Pairwise genetic distances were estimated, using MEGA, version 4.0.2 (Kumar, Tamura & Nei, 2004) and the Kimura two-parameter model (Kimura, 1980). The genetic structure of the populations that revealed special genetic features was investigated by analysis of molecular variance (AMOVA) in ARLEQUIN, version 3.01 (Excoffier, Laval & Schneider, 2005) using AMOVA (Excoffier, Smouse & Quattro, 1992), whereas groups were defined with DNASP, version 4.10.9 (Rozas *et al.*, 2003).

Neighbor-joining (NJ), maximum parsimony (MP) and Bayesian inference (BI) analyses were conducted. The NJ analysis was also carried out, using the Kimura two-parameter model (Kimura, 1980) by MEGA, version 4.0.2 (Kumar *et al.*, 2004). Confidence in the nodes was evaluated by 1000 bootstrap replicates (Felsenstein, 1985). The MP analysis was performed using PAUP*, version 4.0b10 (Swofford, 2002), with heuristic searches, using stepwise addition of sequences and performing tree-bisection-reconnection branch swapping (Swofford *et al.*, 1996). Confidence in the nodes was evaluated by 1000 bootstrap replicates (Felsenstein, 1985). For both trees, a clade was considered as well supported, when the bootstrap support was higher than 80%. BI analysis was performed with MrBayes, version 3.1 (Ronquist & Huelsenbeck, 2003) applying the parameters of the substitution model, suggested by MODELTEST, version 3.7 (Posada & Crandall, 1998), according to the Akaike information criterion (Akaike, 1974). The number of generations was set to 1×10^6 . Support for nodes was assessed with the posterior probabilities of reconstructed clades as estimated by MrBayes, version 3.1 (Ronquist & Huelsenbeck, 2003). A clade was considered as well supported, when the posterior probability was higher than 0.95.

RESULTS

KARYOLOGICAL STUDY

The karyological study of the 53 individuals of *Acomys minous* revealed the existence of three homozygous karyotypes with $2n = 42$, 40 and 38, respectively, and stable FNa = 66, FN = 68 (Table 1). The G-banding staining technique showed that these karyotypes are



Figure 2. G-banded karyograms of *Acomys minous* (chromosomes sorted with decreasing size), A, B, homozygous males with $2n = 42$ and $2n = 40$, respectively, the latter carrying additionally Rb(17.19). C, a homozygous female with $2n = 38$ and the Rb(15.16) and Rb(17.19). D, E, hybrid males with $2n = 39$ and $2n = 41$, respectively, heterozygous for Rb(15.16). F, a hybrid male with $2n = 41$, heterozygous for Rb(17.19).

closely related because they show complete homology for 13 biarmed and three acrocentric autosomal pairs, as well as for the sex chromosomes. In particular, eight individuals (four males, four females) from five localities of Crete demonstrated $2n = 42$ and were characterized by 13 biarmed (mostly meta- or submetacentric) and seven acrocentric autosomal pairs (Fig. 2A). Twenty-one individuals (15 males, six females) from seven localities were characterized by

$2n = 40$ and carried additionally Rb(17.19), fixed in homozygosity, following an Rb fusion of the respective acrocentrics (Fig. 2B). Moreover, four individuals (three males, one female) from three localities displayed $2n = 38$. Compared to individuals with $2n = 40$, these possessed an additional biarmed autosomal pair [i.e. Rb(15.16)], which is the largest of the complement, as a result of the Rb fusion between the respective acrocentric autosomes of the latter (Fig. 2C).

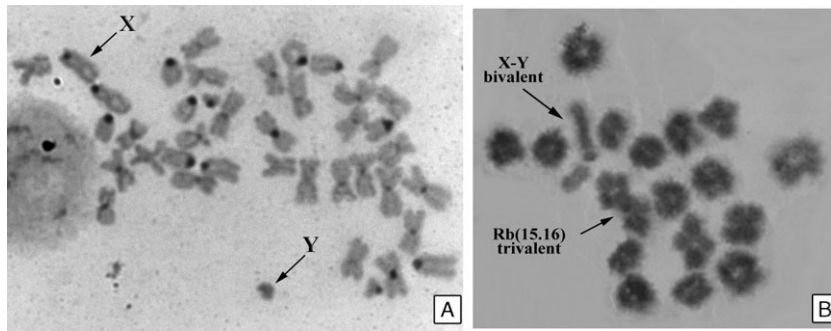


Figure 3. A, C-banded metaphase spread of a male *Acomys minous* with $2n = 42$ (sex chromosomes indicated by arrows). B, meiotic metaphase spread of a male *A. minous*, heterozygous for Rb(15.16) ($2n = 39$). The synaptic X-Y bivalent, as well as the trivalent, as a result of the heterozygous Rb condition are indicated.

Furthermore, 20 of the studied individuals (37.7% of the sample), among them, several females in pregnancy, were heterozygous for one biarmed chromosome. More specifically, nine individuals (six males, three females) from three localities were characterized by $2n = 39$, as a result of hybridization between individuals with $2n = 38$ and $2n = 40$ that led to the appearance of Rb(15.16) in heterozygosity (Fig. 2D). Interestingly, two individuals (one male, one female) from Linoperamata (Loc. 4; Fig. 1, Table 1) were also heterozygous for Rb(15.16) but had $2n = 41$, as a result of successive hybridization events that started between individuals with $2n = 38$ and $2n = 42$. Finally, nine individuals (five males, four females) from five localities demonstrated $2n = 41$ but were heterozygous for Rb(17.19), as a result of hybridization between individuals with $2n = 40$ and $2n = 42$ (Fig. 2F). In all karyotypes, the X chromosome was a very large acrocentric, almost similar in size to the largest acrocentric autosomal pair and sometimes appeared with a barely visible small pair of arms. In some females, these tiny arms appeared in heterozygosity. On the other hand, the Y chromosome appeared as a small acrocentric, equal in size to the smallest autosomal pair of the complement.

The application of the C-banding staining technique showed that constitutive heterochromatin is restricted to the pericentromeric region of all acrocentric and most metacentric autosomes, as well as of the X chromosome. The small arms of the X chromosome, wherever present, appeared euchromatic, whereas the Y chromosome was uniformly, yet faintly heterochromatic (Fig. 3A). It is notable that the biarmed chromosomes presented significantly less prominent pericentromeric heterochromatin or none at all, compared to acrocentric autosomes.

Meiotic preparations in males of *A. minous* revealed that all chromosomes formed bivalents during meiosis, including the X and Y chromosomes, which appeared synaptic, whereas all hybrid males dis-

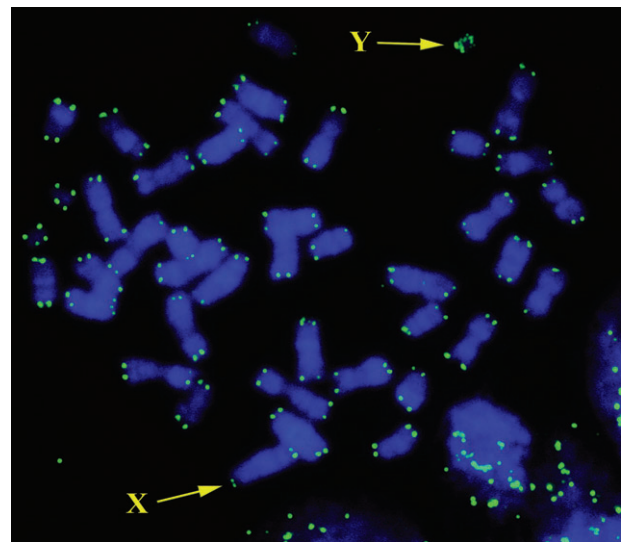


Figure 4. Fluorescent *in situ* hybridization with the telomeric probe, in a metaphase of a male individual of *Acomys minous* with $2n = 42$. The chromosomes are stained with 4'-6-diamidino-2-phenylindole and arrows indicate the sex chromosomes.

played one trivalent, corresponding to the single heterozygous Rb fusion in each case (Fig. 3B).

Finally, the *in situ* hybridization with the telomeric probe provided a clear pattern at both ends of all chromosomes, as expected, whereas no interstitial telomeric sites (ITS) were detected. Notably, Y chromosome appeared to have the most intense signal, compared to all the other chromosomes of the complement (Fig. 4).

MOLECULAR STUDY

The sequencing of the *cytb* gene studied produced an alignment of 1073 bp. The number of variable sites was 270, whereas only 188 of them were

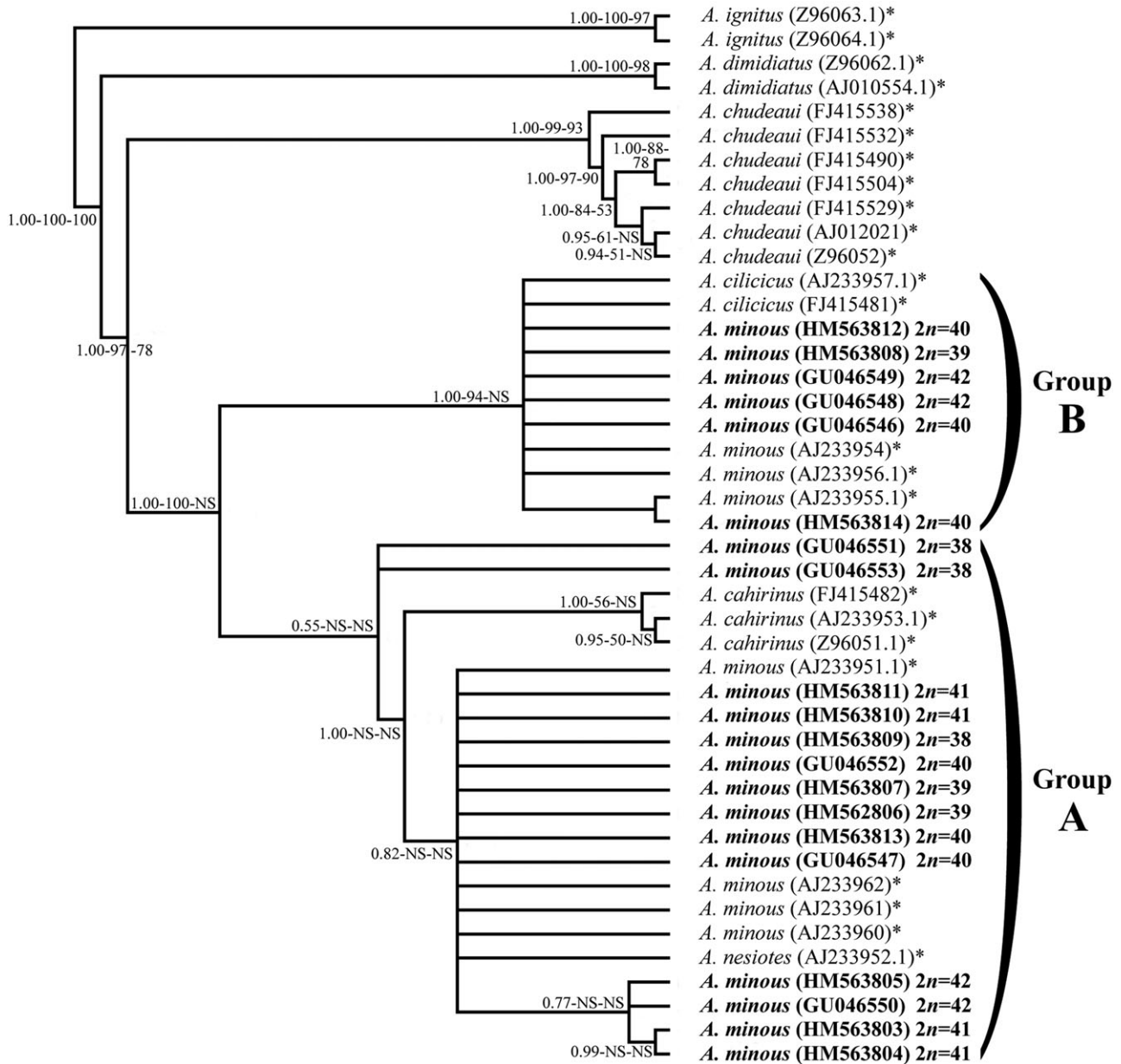


Figure 5. Phylogenetic tree constructed by Bayesian inference for the cytochrome *b* (cytb) gene. Numbers in nodes represent Bayesian posterior probabilities, maximum parsimony and Neighbour-joining bootstrap supports, respectively. Parentheses, refer to GenBank accession numbers. Sequences in bold refer to the present study, whereas those marked with an asterisk are derived from previous studies (Barome *et al.*, 1998, 2001a, b; Nicolas *et al.*, 2009). For *Acomys minous* specimens from the present study, the diploid chromosome number is also provided. N.S., not supported.

parsimony informative. In 53 instances, the nucleotide substitution led to amino acid replacement. The highest divergence value (15.7%) was observed between *Acomys ignitus* and all the others. The average pairwise genetic distances were 4.5% (Kimura two-parameter; Kimura, 1980).

The best-fit model, selected by MODELTEST, version 3.7 (Posada & Crandall, 1998), was GTR+I

(base frequencies: A = 0.3043, C = 0.3151, G = 0.1308, T = 0.2497, proportion of invariable sites = 0.6705 and equal rates for all sites).

The phylogenetic trees obtained by the NJ, MP and BI analyses demonstrated similar topologies (i.e. four major clades, supported by high bootstrap values/posterior probability); therefore, only the BI tree is presented here (Fig. 5): (1) *A. ignitus*, (2) *A. dimidia-*

Table 2. Mean genetic distance (Kimura two-parameter) between *Acomys* clades

	<i>Acomys ignitus</i>	<i>Acomys dimidiatus</i>	<i>Acomys chudeaui</i>	<i>Acomys cahirinus</i>	<i>Acomys cilicicus</i>	<i>Acomys nesiotes</i>	<i>Acomys minous</i> 'A'	<i>Acomys minous</i> 'B'
<i>Acomys ignitus</i>	–							
<i>Acomys dimidiatus</i>	0.077	–						
<i>Acomys chudeaui</i>	0.072	0.077	–					
<i>Acomys cahirinus</i>	0.084	0.087	0.069	–				
<i>Acomys cilicicus</i>	0.088	0.095	0.067	0.014	–			
<i>Acomys nesiotes</i>	0.097	0.091	0.072	0.010	0.010	–		
<i>Acomys minous</i> 'A'	0.091	0.086	0.067	0.007	0.007	0.003	–	
<i>Acomys minous</i> 'B'	0.088	0.095	0.067	0.014	0.000	0.010	0.007	–

A. minous 'A' and *A. minous* 'B' refer to the cytochrome *b* distinction of *A. minous* populations into Group 'A' and Group 'B' lineages *sensu* Barome *et al.* (2001a) and the present study.

tus, (3) *A. chudeaui* (including specimens formerly attributed to *A. airensis*), and (4) *cahirinus* group (including the taxa *A. minous*, *A. cahirinus*, *A. cilicicus*, and *A. nesiotes*). In the *cahirinus* group, *cytb* shows a polytomy, the branches of which include Group 'A' of *A. minous sensu* Barome *et al.* (2001a) grouped with *A. nesiotes* and *A. cahirinus* and Group 'B' of *A. minous*, grouped with *A. cilicicus*. The mean genetic distance (Kimura two-parameter) between the taxa of the *cahirinus* group and the species *A. airensis*, *A. dimidiatus*, and *A. ignitus* is given in Table 2. The genetic structure as revealed by AMOVA among the above four clades demonstrates that most of the genetic variability was observed among groups (82.48%), and a small proportion of the variability was observed among populations within groups (0.23%) or within populations (17.29%).

DISCUSSION

The present study revealed the existence of a new chromosomal population for *A. minous* with $2n = 42$. For this population, as well as for the one with $2n = 40$, G-banding analysis was performed for the first time. Therefore, *A. minous* consists of at least three Rb populations, with $2n = 42$, 40 and 38 chromosomes, respectively (Matthey, 1963; Kunze *et al.*, 1999a; present study). The Rb population with $2n = 42$ appears to have led to the formation of the two populations with $2n = 40$ and $2n = 38$, through the successive fixation in homozygosity of the two additional Rb fusions [i.e. of Rb(17.19) and, at a later stage, of Rb(15.16)]. Regarding the Rb population with $2n = 38$, the results of the present study agree with those of Kunze *et al.* (1999a) (i.e. FN = 68), rather than those pertaining to Race I *sensu* Matthey (1963) (i.e. FN = 66). The fact that *Acomys nesiotes* and the Rb population with $2n = 38$ of *A. minous* have identical karyotypes (Zima *et al.*, 1999; Kunze *et al.*,

1999a), and that *A. cahirinus* and *A. cilicicus* just possess an additional Rb fusion between acrocentrics 14 and 18 (Fig. 2) (El Ashmawy, 1990; Macholán *et al.*, 1995; Kunze *et al.*, 1999a), indicate that the Rb population with $2n = 42$ of *A. minous* is the ancestral karyotypic form for the whole *cahirinus* group.

On the basis of the chromosomal constitution of the numerous hybrids in the present study sample, it appears that fertile individuals with simple chromosomal heterozygosities readily occur at contact areas among all three *A. minous* Rb populations. This is a feature that has also been demonstrated at contact areas between Rb chromosomal populations of other murid taxa, such as *Mus musculus domesticus* (Scriven, 1992; Hauffe & Searle, 1993). On the other hand, the synaptic behavior of the X-Y bivalent in *A. minous* during meiosis has also been recorded for *A. dimidiatus* (Wahrman & Goitein, 1972) and the same is expected to be true for the other members of the *cahirinus* group.

With regard to the X chromosome morphology, the tiny arms must be an established polymorphism in the *cahirinus* group because they were occasionally present in our *A. minous* sample and have also been mentioned in *A. cilicicus* (Macholán *et al.*, 1995; Arslan *et al.*, 2008), in *A. nesiotes* (Zima *et al.*, 1999), and in some individuals of *A. cahirinus* (Volobouev *et al.*, 1996b).

The distribution of heterochromatin in the karyotype of the studied sample agrees with that from the other species of the *cahirinus* group with small exceptions (i.e. the existence of interstitial X chromosome heterochromatic bands in *A. cahirinus* from Ethiopia, the absence of heterochromatin in the Y chromosome of *A. nesiotes* and the fully heterochromatic, tiny arms of the X chromosomes in *A. cilicicus* and *A. nesiotes*) (Sokolov *et al.*, 1993; Macholán *et al.*, 1995; Zima *et al.*, 1999).

The telomeric sequences (TTAGGG) $_n$ are normally located in long tandem repeats at the end of all chromosomes of the vertebrates, whereas they can also appear in ITS or pericentromeric regions. Such findings may constitute remnants of chromosomal rearrangements, such as tandem fusions, Rb fusions or pericentric inversions (Meyne *et al.*, 1990; Garagna *et al.*, 1997; Castiglia, Gornung & Corti, 2002; Castiglia, Makundi & Corti, 2007). Volobouev *et al.* (1996a), proposed that the chromosomal evolution of *Acomys* sp. ($2n = 68$, FN = 68), and actually of the whole *cahirinus-dimidiatus* group, occurred exclusively through the fixation of Rb fusions. The absence of ITS in our sample suggests that no interstitial telomeric sequences remained after the chromosomal rearrangements that occurred during the chromosome evolution of *A. minous*. Indeed, complete loss of telomeric sequences, following an Rb fusion (visualized in our sample through the comparatively smaller amount of pericentromeric heterochromatin in Rb chromosomes), has been demonstrated to occur in other murid taxa (e.g. *M. m. domesticus*). A fine scale analysis of the pericentromeric heterochromatin in the Rb-evolving *Acomys* lineages will reveal whether a potential reorganization of the pericentromeric regions, leading to their increased homology in non-homologous chromosomes, may have facilitated the formation of Rb fusions in the *Acomys* species with low diploid chromosome number (Garagna *et al.*, 1995; Nanda *et al.*, 1995), even though there are indications that the sequences and size of the pericentromeric satellite DNA of *Acomys* species with high and low diploid chromosome numbers are rather similar (Kunze *et al.*, 1999b).

The new data can provide an insight with regard to the chromosomal constitution of the *Acomys* populations that colonized Crete and the colonization process that was followed. It is generally accepted that the colonization of Crete is a rather recent event, because fossil *Acomys* records appear to be lacking from Crete (Dieterlen, 1978). The significant chromosomal similarity between *A. cahirinus* and *A. minous* and the northwards expansion route that the genus appears to have followed from central Africa, during its evolution (Barome *et al.*, 2000), suggest that a predecessor of *A. cahirinus*, perhaps with many Rb chromosomes fixed in its karyotype, colonized the island, taking advantage of human travelling routes during Antiquity, thus giving rise to *A. minous* (Kunze *et al.*, 1999a; Barome *et al.*, 2000, 2001a).

Specifically, it could be true that *Acomys* populations with at least two of the karyotypes that characterize *A. minous* today (i.e. $2n = 42$ and $2n = 38$) could have existed in North Africa as established and geographically distinct chromosomal populations that independently colonized Crete. Reaching the island,

they may have come in parapatric contact and locally given rise to the population with $2n = 40$, through zonal raiation (Searle, 1993). Of course, it cannot be excluded that a chromosomally polymorphic population with $2n = 42 - 2n = 38$ already pre-existed in North Africa and colonized Crete, bringing along this polymorphism. To further clarify this, it would be very beneficial to karyologically screen more populations in other regions within the distribution range of the *cahirinus* group in North Africa, Cyprus, and Anatolia, to determine whether the chromosomal polymorphism, currently known only for *A. minous*, is a trait of the whole group. Also the study of the chromosomal evolution and dispersion routes in the group may further benefit from the application of banding chromosomal techniques and the molecular investigation of two, otherwise poorly studied, North African *Acomys* populations. These populations are distributed in geographically intermediate positions between *A. cahirinus* and *A. chudeaui sensu* Nicolas *et al.* (2009), two species with almost similar diploid chromosome numbers but extensive monobrachial homologies.

The first population, known from central Morocco has been attributed to *A. chudeaui* and conventional staining has shown a karyotype of $2n = 40$ but FN = 70 instead of FN = 68. The difference in FN results from a presumably submetacentric, instead of the typical acrocentric X chromosome (Benazzou, 1983), which could nevertheless be the result of misidentification. On the basis of this meager chromosomal data (no karyotype photograph has ever been provided), this population could be attributed to *A. chudeaui* or the *cahirinus* group, or even constitute a new species altogether; however, this question remains open until new data become available (Nicolas *et al.*, 2009).

The second population that needs to be studied has been attributed to *A. seurati* and is distributed in South Algeria. On the basis of conventional staining, its karyotype appears to be characterized by $2n = 38$, FN = 68 (Matthey & Baccar, 1967). Nicolas *et al.* (2009) expressed the necessity to test the validity of this species by means of a molecular study, and we consider that a banding chromosomal study would prove equally helpful. The current data suggest that this population is closer to the *cahirinus* group rather than to *A. chudeaui* because $2n = 38$ has only been described in the former.

In any case, the small karyotypic differences between the taxa of the *cahirinus* group, as a result of Rb fusions, are generally not considered sufficient to lead to disruption of gene flow and independent divergence of the populations in question (Volobouev *et al.*, 2002). Indeed, various molecular studies on species of the genus *Acomys*, mostly based on the examination of *cytb* sequences, have revealed that the taxa of the

cahirinus group are not clearly distinguishable among each other. The maximum displayed divergence among these sequences in the *cahirinus* group (1.6%) is equivalent to the intraspecific diversity observed in other *Acomys* species (Barome *et al.*, 2001a). Our *cytb* study completely supports the results of such previous studies. In particular, the very low genetic diversity within the *cahirinus* complex was verified, along with the appearance of two distinct maternal lineages within this complex (i.e. Group 'A' of *A. minous*, *A. cahirinus*, and *A. nesiotus* on one hand and Group 'B' of *A. minous* and *A. cilicicus* on the other). In this context, *A. minous* appears paraphyletic to the other taxa of the *cahirinus* group, as has been previously suggested (Barome *et al.*, 2001a). In addition, the results of the present study show that no correlation exists between the chromosomal constitution of the sequenced individuals and the *cytb* clade they are placed into (Fig. 5). Therefore, it is further supported that the divergence of *cytb* sequences took place much earlier than the colonization event of Crete by *Acomys*. This means that the colonizing population, arriving at a single or multiple waves in Crete, already possessed this *cytb* variation, which must have also preceded by much the chromosomal evolution that has taken place in the populations of the *cahirinus* group (0.4 Mya versus 40,000 years ago) (Barome *et al.*, 2001a).

Overall, the results obtained in the present study corroborate the emerging opinion, supported by various approaches (e.g. cytogenetic, molecular, etc.) that the four taxa of the *cahirinus* group constitute a single species, based on their very close molecular distance, recent chromosomal diversification, and low karyotypic dissimilarities, which may not hamper the production of viable and fertile hybrids. In this context, it is reasonable to consider the forms '*cahirinus*', '*cilicicus*', '*minous*' and '*nesiotus*' as subspecies of *A. cahirinus* (Barome *et al.*, 2001a; Volobouev *et al.*, 2002; Kryštufek & Vohralík, 2009). The further application of cytogenetic and mtDNA studies in other, little known populations of the *cahirinus* group, as suggested above, as well as the use of additional, informative approaches (i.e. microsatellite and histocompatibility complex gene analyses) may reinforce this proposal. In any case, one should keep in mind that, because of their geographical isolation, the populations of the proposed subspecies may independently diversify to eventually become distinct species in the future.

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